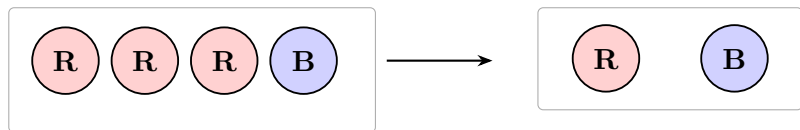


For random draws, use the same bag. Every counter has the same chance each time; put it back and mix after each draw.



Red, then blue makes this pair. Keep the order.

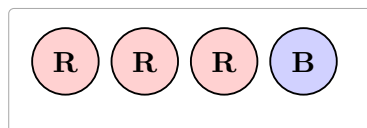
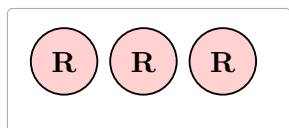
Problem 1: Choose draws yourself to make every different color pair. Which pairs disappear if you take the blue counter out of the bag?

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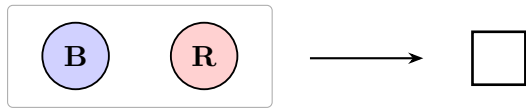
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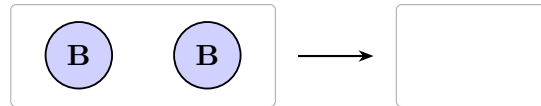
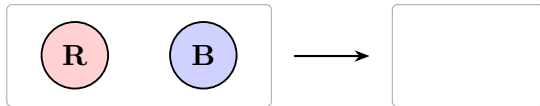
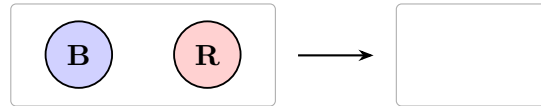
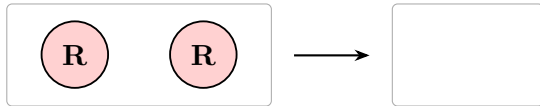


Use three red counters and one blue again. A rule gives each color pair a square, a circle, or no shape. The same color pair always gets the same answer.



A sample rule: BR gives a square.

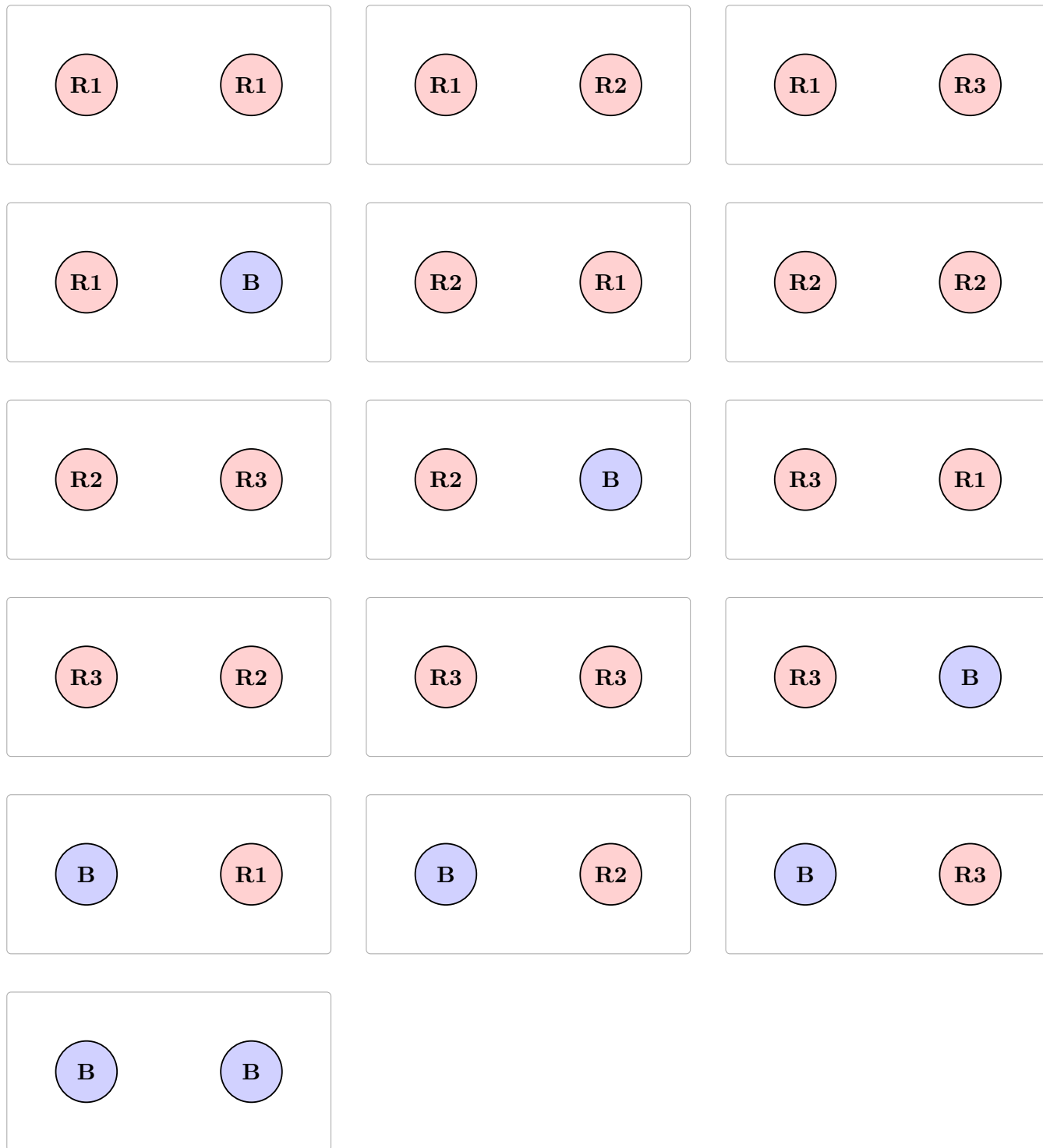
Problem 2: Make a rule that sometimes gives each shape and gives them the same chance. Try it on six random pairs.



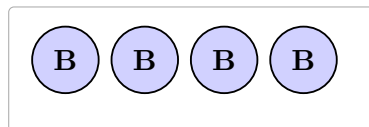
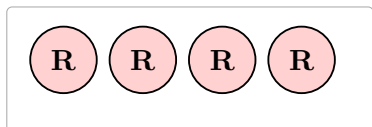
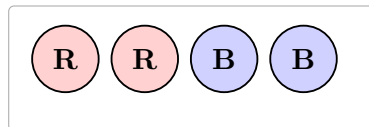
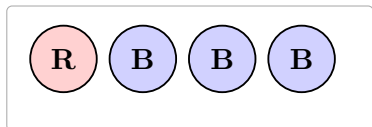
Use a cross to record no shape.

The red counters have marks R1, R2, R3. Each marked pair below has the same chance.

Problem 3: Does your rule give each shape to the same number of pictures? Find every rule that does and sometimes gives each shape.



Problem 4: Keep your rule for these bags. Which can give each shape, and do the shapes have the same chance?



Problem 5: Choose a four-counter bag that makes shapes come out as often as possible. Could any four-counter bag give a shape on every pair?

Problem 6: Choose pairs and draw six-pair stories using your rule for each target.
Can six random pairs force a shape to appear?

no shapes

→	→	→
→	→	→

exactly one square and no circles

→	→	→
→	→	→

exactly three circles and no squares

→	→	→
→	→	→